

Scalability & Future Plan

Date	2 October 2026
Team ID	SWTID-2026-1665
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

EduGenie is currently a prototype that serves a small group of learners. This document explains how the platform can grow to support many more students, richer study material and new learning features, and sets out the team's plan for the stages that follow.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Every request depends on the Google Gemini API and its quota	Answers may slow down or fail when the quota is reached or the service is unavailable	High
2	Quiz scores and notes are kept in a small, simple store	Long-term progress tracking becomes hard as the number of learners grows	High
3	Content is mainly in English and based on typed text	Students who prefer regional languages, voice input or uploaded files are not fully served	Medium
4	No learner accounts or personal dashboards yet	Study history cannot be saved across devices or shared with teachers	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	Traffic Handling	Single web instance serving a small group	Host on auto-scaling cloud services so more instances start automatically during busy study hours
2	Learner Data	Lightweight storage for quizzes and notes	Move to a managed cloud database with regular backups and per-learner records
3	AI Requests	Each question is sent to Gemini directly	Add a request queue, reuse answers to common questions and track usage against the quota

4	Response Speed	Full answers are returned after generation finishes	Stream answers as they are generated and keep frequent study topics ready in a cache
5	Access and Privacy	API keys held in environment variables	Add sign-in, role-based access for students and teachers, and encrypted storage of learner data

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Polish the explanation, summary and quiz screens and fix issues found in user testing	1–2 Months	Smoother, more reliable everyday use
Phase 2	Student accounts, saved study history and a progress dashboard	3–4 Months	Learners can continue from any device
Phase 3	Upload of notes and PDFs, voice questions and adaptive quizzes that follow weak topics	5–6 Months	More personal and flexible learning
Phase 4	Regional language support, teacher view, mobile app and full cloud deployment	6–12 Months	Reach more students and classrooms